

Robust Regression

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Introduction

Robust regression estimators down-weight observations with large residuals to reduce the influence of outliers and heavy-tailed errors. Where ordinary least squares minimises the squared residual — making a single large residual contribute as much as many moderate ones — robust methods cap or smoothly bound the loss function so that extreme observations cannot dominate the fit. Two main families coexist: M-estimators (Huber, bisquare) and MM-estimators that combine a high-breakdown initial fit with a high-efficiency M-step.

Prerequisites

A working understanding of OLS, residual diagnostics, leverage and influence, and the breakdown-vs-efficiency trade-off in robust statistics.

Theory

M-estimators minimise $\sum \rho(r_i/s)$ where ρ is a robust loss function: quadratic for small residuals (where it acts like OLS) and bounded or transitioning to linear for large residuals (capping the influence of outliers). Huber's ρ transitions at 1.345 SDs; bisquare's ρ is exactly zero beyond a cutoff, eliminating large outliers entirely.

MM-estimators combine a high-breakdown initial fit (S-estimator with breakdown 50 %, the maximum possible) with a high-efficiency M-estimator using the S-estimate as starting point. The result has breakdown 50 % and 95 % efficiency at Normal errors — the modern default for robust linear regression.

Assumptions

Outliers in the response variable; for outliers in predictors (high-leverage points), the MM-estimator still resists but the breakdown safeguard is weaker. Errors symmetric or near-symmetric; for severely skewed errors, transformation may be preferable to robust regression.

R Implementation

```
library(MASS); library(robustbase)

# Simulated data with 10% outliers
set.seed(2026)
x <- rnorm(100)
y <- 2 + 1.5 * x + rnorm(100)
y[sample(100, 10)] <- y[sample(100, 10)] + rnorm(10, 0, 10)

fit_ols <- lm(y ~ x)
fit_rlm <- rlm(y ~ x) # Huber M-estimator
fit_mm <- lmrob(y ~ x, method = "MM") # MM-estimator

rbind(ols = coef(fit_ols), rlm = coef(fit_rlm), mm = coef(fit_mm))
```

Output & Results

OLS coefficients are pulled by the contamination; both robust estimators recover values close to the data-generating slope. The MM-estimator typically delivers tighter standard errors than the Huber M-estimator at the same level of robustness, reflecting its better efficiency.

Interpretation

A reporting sentence: “MM-regression gave slope 1.48 (SE 0.11), close to the true 1.5; OLS on the same 10 %-contaminated data produced 1.35 (SE 0.16), pulled toward zero by the outliers. Reporting both fits illustrates the robustness benefit and the cost of ignoring outliers.” Pair OLS with a robust fit when outliers are suspected; large discrepancies signal the influential observations need investigation.

Practical Tips

- `robustbase::lmrob()` with `method = "MM"` is the modern default for robust linear regression; superior to Huber `MASS::rlm()` in efficiency-breakdown trade-off.
- Always report both OLS and robust fits; large discrepancies signal outlier influence and motivate investigation of the contaminated points.
- Robust regression is a fix for outlier influence, not a substitute for understanding why outliers occurred — investigate first, robustify second.
- For outliers in predictor space (high-leverage points), the MM-estimator still resists but the safeguard is reduced; combine with leverage-aware diagnostics.
- Robust SEs (sandwich variance) address heteroscedasticity, not outliers; the two are distinct problems requiring distinct fixes.
- For GLMs, `robustbase::glmrob()` provides analogous robust estimators for binomial, Poisson, and gamma regression.

R Packages Used

`MASS::rlm()` for Huber M-estimators; `robustbase::lmrob()` for MM-estimators with high-breakdown initialisation; `robustbase::glmrob()` for robust GLMs; `quantreg` for quantile regression as a complementary robust alternative; `WRS2` for Wilcoxon’s robust ANOVA and regression analogues.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook’s distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (`emmeans`)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with `mgcv`
- Non-linear regression with `nls`

- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI